

Online Agriculture Products Store

Q1. Identify Business Process Model for Online Agriculture Store – (Goal, Inputs, Resources, Outputs, Activities, Value created to the end Customer)

Goal: Enable farmers/customers to browse, purchase, and receive agriculture products online, streamlining supply chain from supplier to end-user.

Inputs:

- Customer search queries: Product name and category like seeds/fertilizers
- Product catalog data (codes, prices, stock levels).
- Customer details (login, address, payment info)
- Farmer needs: Crop type, product quantity, location, and preferences.
- Manufacturer data: Product details (description, price, stock), images, and certifications.
- Transaction info: Payment methods and delivery addresses.

Resources

- Online portal/system (admin-maintained catalog): Online web/mobile platform with user-friendly interface.
- Database for inventory, orders: Secure database, payment gateway, and inventory system.
- Staff (admin for updates, delivery personnel), Logistics partners for rural delivery and customer support team.

Outputs

- Order confirmation/receipt: Confirmed orders with tracking details.
- Updated inventory records: Invoices, feedback records, and usage analytics for manufacturers.
- Delivery shipment: Delivered products to farmer locations.

Activities

Customer

- Customer searches/logs in and browses catalog.
- Adds items to cart, checks availability.
- Places order, provides details/payment.
- System validates, confirms, updates stock; notifies via email/SMS.
- Admin processes fulfillment/shipment.

Manufacturer

- Manufacturers register and upload/manage product listings.
- Updates the inventory with product availability
- Processes the order fulfillment

Value to End Customer

Convenience of anytime access, competitive pricing, stock visibility, hassle-free transactions/delivery—saving time/energy vs. traditional markets.

Q2. Mr Karthik is doing SWOT analysis before he accepts this project. What Aspects he Should consider as Strengths, as Weaknesses, as Opportunity and as Threats.

Strengths • Ready talent pool: Experienced team having PM, Java devs, admins and BA support enable quick ramp-up • Strong sponsorship: CSR backed • Noble mission: Addresses genuine farmer pain points • Agile/SDLC flexibility for quick delivery	Weakness • Potential domain gap in agriculture (supply chain perishables vs. retail) • Scope creep risk from changing farmer needs/seasonal demands. • Fixed constraints: 18 month timeline and 2 crs budget may strain for full mobile/web app with logistics integration, risking delays if scope expands. • Rural tech barriers: Farmers' low digital literacy demands heavy UX investment, potentially increasing development costs.
Opportunities • Growing rural digital adoption (e-commerce for seeds/fertilizers) addressing market gap of booming agri-e-commerce • Partnerships with suppliers/farmers for real-time inventory. • ROI via scalable online platform, market expansion (e.g., mobile app) • CSR angle attracts govt. subsidies or expansions into advisory services.	Threats • Competitors (existing agri-apps/markets), negative stakeholders (traditional sellers) & new entrants face market saturation. • Regulatory changes (agri policies, data privacy for payments). • External risks like supply disruptions, low rural internet/tech literacy (Poor rural internet/logistics) • Scope creep from stakeholders or integration issues (payments/delivery) could overrun budget/timeline.

Q3. Mr Karthik is trying to do feasibility study on doing this project in Technology (Java), Please help him with points (HW SW Trained Resources Budget Time frame) to consider in feasibility Study.

Hardware Considerations

Servers for development/UAT/production (e.g., scalable cloud instances for inventory/orders), high-availability setups for concurrent farmer users, and peripherals for testing (mobile/responsive agri-apps).

- Servers/cloud infrastructure (e.g., AWS/EC2) for hosting scalable app handling farmer traffic and inventory.
- Developer workstations (laptops or PCs) and testing devices (smartphones/tablets for mobile UI).
- Network setup for secure data transfer; ensure rural delivery integration doesn't overload systems.

Software Considerations

Java tech stack (Spring Boot for microservices, Hibernate for DB), e-commerce frameworks (catalog/payment APIs), tools like Jenkins/Docker for CI/CD, and compatibility with agri-specific needs (e.g., real-time stock).

- Java backend (Spring Boot for APIs, Hibernate for ORM) with MySQL/PostgreSQL database.
- Frontend: React/Angular for web, React Native for mobile cross-platform.
- Integrations: Payment gateways (Razorpay), logistics APIs; security (SSL, JWT).

Trained Resources Considerations

Java developers (backend/frontend), BAs for UML modeling (12-16% team allocation), testers for UAT, and domain SMEs; assess availability/skills via RACI matrix to avoid gaps.

- Leverage existing team: Senior Java dev (Juhi), 4 Java devs, PM (Vandanam), DB admin (John), network admin (Mike), testers (Jason, Alekya).
- Assess gaps in e-commerce/microservices; plan 1-2 weeks training if needed for agri-specific features.

Budget

Licensing (Java open-source base but paid tools like IDEs), cloud hosting, team salaries (e.g., fixed-bid vs. billing rates), contingency for scope creep (15% SDLC BA time allocation).

- Allocate ~40% to resources/salaries, 30% SW licenses/cloud, 20% HW/infra, 10% testing/tools—fits within 2 Crores INR.
 - Personnel Costs (~₹80 Lakhs, 40%): Covers salaries for 12-15 member team (PM, devs, admins, testers, BA) at avg. ₹8-12L/year onshore rates. Includes training (₹2-3L) for Java/Spring Boot and domain skills; hiring buffers if needed. Phased: 50% dev phase, 30% testing, 20% planning/UAT.

- Software & Tools (~₹40 Lakhs, 20%): Java ecosystem (IDEs like IntelliJ ~₹1L/team), frameworks (free Spring Boot), DB licenses (MySQL/PostgreSQL free/open), payment gateways (Razorpay setup ₹2-5L), testing tools (Selenium/JUnit). Cloud dev/test subscriptions (AWS free tier initially).
- Hardware & Infrastructure (~₹30 Lakhs, 15%): Developer laptops/servers (₹10-15L), cloud infra (AWS/EC2 ₹10-15L/year for scalable hosting/inventory). Testing devices (mobiles ₹3-5L); network/security setup. Optimize with BYOL for 40% savings.
- Cloud & Operations (~₹30 Lakhs, 15%): Ongoing AWS/GCP hosting (₹15-20L/year post-launch), domain/SSL (₹1L), logistics/payment transaction fees (₹5-10L buffer). Includes maintenance/support for rural scalability.
- Testing, Marketing & Contingency (~₹20 Lakhs, 10%): QA tools/UAT (₹5L), beta testing with farmers (₹3L), marketing/onboarding (₹5L), 10% contingency (₹2L) for scope changes/regulatory tweaks
- Monitor overruns from cloud scaling or custom integrations; Java's maturity keeps costs low vs. newer stacks.

Time Frame Considerations

Total duration typically spans 9-12 months for a medium-large project (500-1000+ man-hours), with BA phase consuming 15% of total SDLC time (e.g., 2 months in a 1-year project). Gantt charts track weekly milestones; no overlap allowed, so delays (e.g., scope creep) cascade; reviews at phase ends determine continuation.

- Requirements Gathering: 4-6 weeks (BA elicits via interviews/JAD, produces BRD/FRD).
- Requirements Analysis: 2-4 weeks (UML modeling, RTM creation).
- Design: 6-8 weeks (Architecture, HDD/DDD for Java stack).
- Development/Coding: 8-12 weeks (Java implementation of catalog/orders).
- Testing: 6-8 weeks (Unit, integration, system, UAT).
- Deployment: 2-4 weeks (Dev-to-UAT-to-Prod servers).
- Maintenance: Post-go-live (ongoing)

Q4. Mr Karthik must submit Gap Analysis to Mr Henry to convince to initiate this project. What points (compare AS-IS existing process with TO-BE future Process) to showcase in the GAP Analysis

AS-IS Process (Current Manual System)

Farmers visit physical stores or middlemen for seeds/fertilizers; limited inventory visibility causes stockouts; cash-based transactions delay payments; manual record-keeping leads to errors and slow delivery.

TO-BE Process (Proposed Online Platform)

Customers browse/search digital catalog anytime; real-time stock updates prevent shortages; secure online payments with order confirmation; automated order processing triggers faster fulfillment/shipping.

Key Gaps and Improvements

Aspect	AS-IS (Gap)	TO-BE (Improvement)
Accessibility	Physical travel to distant markets; limited stock visibility	Nationwide online catalog; anytime/anywhere browsing via mobile
Time Efficiency	Days/weeks for procurement trips and negotiations	Minutes for search/add-to-cart/checkout; real-time stock updates
Cost	Middlemen inflate prices 20-50%; transport costs high	Direct manufacturer pricing; bulk deals; optimized rural logistics
Quality/Trust	No certifications; pest issues from fakes	Verified products with images/certificates; ratings/reviews
Payment/Delivery	Cash-based, delayed payments; erratic rural delivery	Secure digital payments; GPS-tracked delivery to remote areas
Information	Manual inquiries; no crop-specific advice	Search/filter by crop; order history/analytics
Reporting	Paper ledgers, no analytics	Dashboards for sales/inventory trends

This transformation reduces procurement time by 80%, cuts costs by 30%, and boosts farmer productivity—directly addressing stakeholder pain points for CSR impact.

Q5. List down different risk factors that may be involved (BA Risks And process/Project Risks)

Internal Risks

- Inadequate inventory management systems causing stock discrepancies across warehouses.
- Poor customer data organization leading to inaccurate order fulfillment for farmers.
- Weak internal processes for supplier coordination, resulting in delayed restocking.
- Limited internal team expertise in agriculture domain specifics like seasonal crop cycles.

External Risks

- Seasonal demand fluctuations (e.g., monsoon disruptions in fertilizer purchases).
- Regulatory changes in agricultural subsidies or e-commerce taxation impacting pricing.
- Dependency on third-party logistics providers causing unreliable rural deliveries.
- Market competition from established agri-apps eroding customer acquisition.

BA Risks

- Incomplete capture of farmer requirements (e.g., vernacular needs, bulk discount rules).
- Misalignment between merchant expectations and stakeholder inputs from suppliers.
- Failure to document non-functional needs like mobile or webpage responsiveness for rural users.
- Scope creep from conflicting inputs between suppliers, middlemen, and end-buyers.
- Inaccurate UML modeling/RTM traceability causing misalignment with dev handoff.

Project Risks

- Extended go-live delays affecting peak-season sales opportunities. Technical scalability failures during peak rural usage spikes.

- Budget overruns from unplanned business customizations (e.g., regional payment methods).
- Low customer adoption due to inadequate training for first-time rural online shoppers.
- Waterfall schedule slippage where early phase delays cascade to deployment.
- Budget overruns from unforeseen Java customizations for real-time inventory.

Q6. Perform stakeholder analysis (RACI Matrix) to find out the key stakeholders who can take Decisions and Who are the influencers

R = Responsible (does the work), A = Accountable (owns outcome), C = Consulted (two-way input), I = Informed (one-way updates)

R= Green, A=Yellow, C= Orange, I=Pink

Activity / Task	Mr. Henry (Sponsor)	Mr. Pandu (Finance)	Mr. Dooku (Coord.)	Peter/K/Ben (Farmers)	Mr. Karthik (Deliv. Head)	Mr. Vandana (PM)	BA	Dev Team	Testers
Approve Project	A	C	C	I	R	C	I	I	I
Define Requirements	A	I	C	C	A	R	R	I	I
Approve Budget Changes	A	A	C	I	C	I	I	I	I
Design System Architecture	I	I	I	C	A	R	C	R	I
Development	I	I	I	I	A	R	C	R	C
UAT (Testing)	A	C	C	R	A	R	C	C	R
Go-Live Approval	A	A	C	C	R	A	I	I	I
Maintenance	I	C	I	C	A	R	C	R	C

Decision-Makers

- Business Owner/Executive Sponsor (Mr. Henry): Accountable for initiation, funding, approvals (signs agreements).

Influencers

- End Users (Farmers/Users): Consulted for requirements; prioritize needs like stock alerts.
- SMEs/Process Experts: Consulted on agri processes; shape feasibility.
- PM/BA: Responsible influencers on execution details.

Q7. Help Mr Karthik to prepare a business case document

1. Why is this project initiated?

Mr. Henry identified remote farmers' procurement difficulties for seeds/fertilizers/pesticides. Project connects farmers directly with manufacturers under 18-month CSR initiative.

2. What are the current problems?

- Farmers travel long distances to buy agri products
- Middlemen inflate prices
- Delayed procurement affects crop cycles
- Manufacturers lack direct farmer feedback
- No centralized product catalog

3. With this project, how many problem, could be solved?

- Online platform eliminates travel
- Direct manufacturer pricing (no middlemen)
- 24-48hr delivery vs weeks
- Real-time inventory/ordering
- Farmer-manufacturer communication

4. What are the resources required?

- 2,950 BA hours (Design:480, Dev:910, Testing:360, UAT:200, Deploy:80)
- AWS infra + Razorpay
- Mr. Karthik (Delivery Head), Mr. Vandanam (PM)
- Peter/Kevin/Ben (farmer reps), Jason/Alekya (QA)
- Mr. Pandu (UAT sign-off)

5. How much organizational change is required to adopt this technology?

The changes will be minimal

- Leverages existing team structure
- New agri-domain knowledge only
- Mr. Karthik retains delivery authority
- Standard SDLC processes apply

6. What is the time frame to recover ROI?

For Mr. Karthik's Online Agriculture Products Store using Waterfall methodology, the ROI recovery time frame is 29 months total (11 months post-launch). With a ₹2 crore (₹200L) budget over 18 months project duration, Waterfall completes development by Month 18 with no revenue during construction. Cumulative net revenues from Month 19 recover the full investment by Month 29, achieving profitability.

Waterfall Timeline (18-Month Duration, ₹2 Cr Budget)

Sequential phases lock ₹200L spend before any revenue generation:

Phase	Duration (Months)	Cost (₹L)	Cumulative Cost
Requirements	1-3	25	25
Design	4-6	35	60
Development	7-13	90	150
Testing	14-17	40	190
Deployment	18	10	200

Zero revenue Months 1-18. Revenue starts Month 19 post-deployment.

Detailed Break-Even Calculation

Cumulative cash flow method tracks net inflows after Month 18:

Month	Revenue (₹L)	Variable Costs (₹L)	Net Cash Flow (₹L)	Running Total (₹L)
1-18	0	200	-200	-200
19	22	5	17	-183
20	35	8	27	-156
21-24	95	22	73	-83
25-28	250	60	190	107
29	320	80	240	40 (break-even)

Payback = Development Time + (Investment ÷ Avg Monthly Net Inflow)

= 18 months + ($\text{₹}200\text{L} \div \text{₹}18\text{L/month}$ post-launch pace)
= 18 + 11.1 months = 29.1 months

7. How to identify stakeholders?

- Interviews - Comprehensive Coverage:
 - Diverse Stakeholder Groups: Farmers, urban retailers, logistics, regulators
 - Unknown Requirements: New e-commerce platform - no existing documentation
 - ₹200L Investment Risk: Must identify ALL impacted parties to avoid scope gaps
- RACI Matrix - Role Clarity + Accountability

Q8. The Committee of Mr. Henry , Mr Pandu , and Mr Dooku and Mr Karthik are having a discussion on Project Development Approach.

Mr Karthik explained to Mr. Henry about SDLC. And four methodologies like Sequential Iterative Evolutionary and Agile. Please share your thoughts and clarity.

Sequential Methodology (Waterfall)

Sequential follows a linear, phase-by-phase approach where each stage completes fully before the next begins. Requirements gathering, system design, development, testing, deployment, and maintenance occur in strict order.

Strengths for Our Project:

- Fixed Rs 2 crore budget provides complete cost certainty
- Eighteen-month timeline suits stakeholder planning cycles
- Comprehensive documentation ideal for regulatory compliance
- Suitable when farmer and retailer requirements remain stable

Challenges:

- No revenue generation during eighteen months of development
- Changes after requirements freeze become extremely costly
- Twenty-nine month total payback period versus eleven months with Agile

Best Fit: Projects with well-defined, stable requirements and regulatory needs.

Iterative Methodology (RUP - Rational Unified Process)

Iterative breaks the project into multiple cycles or iterations, each delivering a working portion of the system. Each iteration includes planning, requirements, design, implementation, and testing.

Key Characteristics:

- Risk-driven iterations focus on highest business value first
- Architecture stabilizes early through elaboration phase
- Continuous refinement based on stakeholder feedback

For Our Agri Platform:

- Could deliver farmer registration module in iteration one
- Retailer ordering system in iteration two
- Payment integration in iteration three
- Progressive revenue generation possible

Trade-off: More complex management than pure sequential approach.

Evolutionary Methodology (Spiral Model)

Evolutionary emphasizes risk analysis at each iteration. Each spiral includes planning, risk assessment, engineering, and evaluation, building progressively more complete versions.

Distinctive Features:

- Explicit risk management focus
- Customer evaluation at end of each spiral
- Suitable for high-risk, large-scale projects

Application Consideration:

- Rs 2 crore investment justifies thorough risk analysis
- Agriculture supply chain uncertainties benefit from spiral approach
- Multiple stakeholder validation reduces market risk

Consideration: Higher management overhead due to risk analysis cycles.

Agile Methodology (Scrum)

Agile delivers working software in short, two to four week sprints. Requirements evolve through continuous feedback from farmers and retailers.

Proven Advantages for E-commerce:

- Revenue generation begins Month 4 versus Month 19 with Waterfall
- Eleven month payback versus twenty-nine months
- Rapid adaptation to changing farmer supply patterns
- Continuous retailer feedback improves platform adoption

Implementation Structure:

- Product Owner: Mr. Karthik defining business priorities
- Scrum Master: Project Manager removing impediments
- Development Team: Cross-functional delivery unit
- Daily stand-ups, sprint reviews, retrospectives

Q9. They discussed models in SDLC like waterfall RUP Spiral and Scrum . You put forth your understanding on these models.

When the APT IT SOLUTIONS company got the project to make this online agriculture product store,there is a difference of opinion between a couple of SMEs and the project team regarding which methodology would be more suitable for this project. SMEs are stressing on using the V model and the project team is leaning more onto the side of waterfall model. As a business analyst, which methodology do you think would be better for this project?

Waterfall would be the better option to proceed with as compared to the V model because of:

- **Budget Predictability - Critical Factor:** V model requires a parallel testing team in each phase which could take a significant budget in every phase whereas the waterfall model; only have one testing phase.
- **Eighteen-Month Timeline Feasibility:** 18 months fits perfectly for sequential development whereas V model would require additional months due to added parallel testing phases.
- **Resource Optimization for Fixed Budget:** V model would require more members for QA team due to additional testing constraints whereas the current setup[of around 4 members for testing team would suffice the requirements for Waterfall model.

- **Budget Protection Controls:** Waterfall model provides [protection for the project to stay within the 2 crore budget. This could be achieved by:
 - Monthly expenditure reviews at phase gates
 - Change requests require committee approval + budget impact analysis
 - 10% contingency reserve only for critical defects
 - Fixed-price subcontractor agreements

Addressing SME V-Model Concerns

- SME Argument: "Better Testing Coverage": Not really, waterfall includes comprehensive testing phase (M14-17)
- SME Concern: "Early Defect Detection": There would be review conducted in each phase which could provide early validation for any available defects without parallel resource overhead.

Risk Assessment: Waterfall vs V-Model

Risk Factor	Waterfall	V-Model	Winner
Budget Overrun	Low	High	Waterfall
Timeline Slip	Medium	High	Waterfall
Testing Coverage	High	High	Tie
Scope Creep	Low	Medium	Waterfall
Resource Peak	Medium	High	Waterfall

Final Recommendation to Committee

Adopt Waterfall methodology immediately because:

- Rs 2 crore fixed budget eliminates V-Model's parallel testing luxury
- Eighteen-month contractual timeline fits sequential phases perfectly
- APT IT Solutions' twelve-member team cannot support V-Model parallelism
- Comprehensive testing phase delivers equivalent quality to V-Model
- Fixed scope contract demands upfront requirements freeze

Waterfall guarantees delivery within Rs 2 crore budget and eighteen-month timeline while providing rigorous quality assurance equivalent to V-Model benefits.

Q10. Write down the differences between waterfall model and V model.

1. Basic Structure and Flow

- Waterfall Model follows a linear, sequential progression from top to bottom: Requirements → System Design → Detailed Design → Coding → Testing → Deployment → Maintenance. Each phase must complete entirely before moving to the next phase.
- V-Model creates a V-shaped structure where development phases on the left side (Requirements → Design → Coding) mirror corresponding testing phases on the right side (Unit Testing → Integration Testing → System Testing → Acceptance Testing). Testing planning occurs parallel to each development phase.

2. Testing Approach and Timing

- Waterfall Model: Testing occurs as a single phase at the end, after all development completes. System testing, integration testing, and acceptance testing happen sequentially after coding finishes.
- V-Model: Testing activities run parallel to development. For each development phase, there's a corresponding test level planned simultaneously. Unit testing plans exist during coding phase, system test plans exist during requirements phase.

3. Flexibility and Change Management

- Waterfall Model: Extremely rigid. Changes after requirements freeze require returning to earlier phases, making modifications costly and time-consuming. Best suited for projects with stable, well-defined requirements.
- V-Model: Slightly more flexible than Waterfall due to early test planning, but still sequential. Changes remain expensive but early verification provides some accommodation before coding begins.

4. Defect Detection and Quality Assurance

- Waterfall Model: Defects discovered late in development cycle during final testing phase. Fixing issues requires rework of completed phases, increasing cost exponentially.
- V-Model: Early defect detection through parallel verification. Each development phase validated immediately by corresponding test planning, reducing defect propagation and late discoveries.

5. Resource Utilization

- Waterfall Model: Sequential resource allocation. Development team works first, then testing team. Peak resource usage occurs during development phase only.
- V-Model: Parallel resource requirements. Development team works while test planning occurs simultaneously. Higher peak resource demand as both development and testing personnel needed concurrently.

6. Cost Structure

- Waterfall Model: Lower overall testing cost as testing consolidated into single phase. Cost of quality lower upfront but potentially higher if major defects found late.
- V-Model: Higher upfront testing cost due to parallel test planning across all phases. Total cost of quality lower as defects caught early, reducing expensive late fixes.

7. Documentation Overhead

- Waterfall Model: Comprehensive phase-wise documentation. Each completed phase produces detailed deliverables before proceeding.
- V-Model: Even more documentation due to test planning documents created parallel to each development phase. Traceability matrices link requirements to specific test cases.

8. Risk Profile

- Waterfall Model: Higher project risk as product not validated until final testing. Major requirements errors discovered after significant investment.
- V-Model: Lower project risk through continuous verification. Each phase validated before proceeding, reducing accumulation of technical debt.

9. Project Suitability

- Waterfall Model: Ideal for small-to-medium projects with clear, stable requirements. Fixed budget and timeline projects. Maintenance-focused applications.
- V-Model: Best for safety-critical systems, medical software, embedded systems where quality assurance paramount. Projects requiring regulatory compliance and traceability.

10. Team Structure Requirements

- Waterfall Model: Traditional team structure. Developers followed by testers. Clear handoffs between phases.
- V-Model: Requires test planning expertise early in project. Testers involved from requirements phase. Higher coordination between development and QA teams.

Q11. As a BA, state your reason for choosing one model for this project.

My recommendation is to proceed with Waterfall methodology as it better aligns with fixed Rs 2 crore budget constraint and twelve-member team structure, while V-Model's quality benefits achievable through rigorous Waterfall testing phase (Months 14-17).

Waterfall Advantages for Rs 2 Crore Budget:

- Sequential testing fits twelve-member team capacity
- Lower peak resource requirements during eighteen-month timeline
- Fixed budget protected through phase gate approvals

V-Model Challenges for Fixed Budget:

- Parallel test planning exceeds team capacity
- Higher testing budget allocation needed
- Resource contention during peak development-testing overlap

Both models sequential in nature, but Waterfall provides superior budget predictability essential for our project's financial constraints.

Q12. The Committee of Mr. Henry, Mr Pandu, and Mr Dooku discussed with Mr Karthik and finalised on the V Model approach (RG, RA, Design, D1, T1, D2, T2, D3, T3, D4, T4 and UAT)

Mr Vandanam is mapped as a PM to this project. He studies this Project and Prepares a Gantt chart with V Model (RG, RA, Design, D1, T1, D2, T2, D3, T3, D4, T4 and UAT) as development process and the Resources are PM, BA, Java Developers, testers, DB Admin, NW Admin.

RG = Requirements Gathering

RA = Requirements Analysis

Design = System/Architecture Design

D1-T1 = Development 1 + Unit Testing 1

D2-T2 = Development 2 + Unit Testing 2

D3-T3 = Development 3 + Unit Testing 3

D4-T4 = Development 4 + Unit Testing 4

UAT = User Acceptance Testing

					Months																	
Phase	Duration	Resources assigned	Start date	Due Date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
RG	3 months	PM, BA	01/01/26	03/31/26																		
RA	3 months	PM, BA	02/01/26	04/30/26																		
Design	3 months	PM, BA, DB Admin	04/01/26	06/30/26																		
D1	2 months	Java Devs (4), DB Admin	07/01/26	08/31/26																		
T1	1 month	Testers (3)	08/15/26	09/14/26																		
D2	1 month	Java Devs (4), NW Admin	09/15/26	10/14/26																		
T2	1 month	Testers (3)	10/15/26	11/14/26																		
D3	1 month	Java Devs (4), DB Admin	11/16/26	12/12/26																		
T3	1 month	Testers (3)	12/14/26	01/13/27																		
D4	1.5 months	Java Devs (4), NW Admin	01/14/27	02/27/27																		
T4	1 month	Testers (3)	03/01/27	03/31/27																		
UAT	3 months	PM, BA, Testers, All	04/01/27	06/30/27																		

Q13. Explain the difference between Fixed Bid and Billing projects

Fixed Bid and Billing (Time & Materials) projects represent two primary contract models in software development. Fixed Bid and Billing (Time & Materials/T&M) projects differ fundamentally in pricing, risk, and flexibility.

1. Pricing Mechanism

- Fixed Bid establishes a single, locked total price upfront (Rs 2 Crores) covering all phases from Requirements Gathering through UAT, regardless of actual hours worked.
- Billing charges based on actual time logged at fixed hourly/daily rates per role (PM, BA, Java Developers, Testers, etc.), plus material costs, invoiced monthly via timesheets.

2.Scope Flexibility

- Fixed Bid requires complete scope definition during Requirements Analysis phase. Changes trigger formal Change Orders needing committee approval (Henry, Pandu, Dooku).
- Billing offers unlimited flexibility—new features or rework during development iterations get billed as actual hours without renegotiation.

3. Risk Distribution

- Fixed Bid transfers 100% execution risk to vendor (Mr. Vandanam/APT IT Solutions). Client pays fixed Rs 2 Crores even if project overruns internally.
- Billing shares risk proportionally—client pays only for actual work, including fixes for issues discovered during testing phases.

4. Cost Predictability

- Fixed Bid provides maximum budget certainty: Mr. Karthik knows maximum exposure = Rs 2 Crores from contract signing.
- Billing creates variable costs requiring ongoing monitoring; actual spend depends on team efficiency and scope evolution.

5. Timeline Control

- Fixed Bid contractually binds vendor to specific delivery dates, with penalties for missing milestones.
- Billing treats timelines as estimates—scope changes naturally extend calendar without contractual breach.

6. Payment Structure

- Fixed Bid: Milestone-based payments tied to phase completions (RG, Design, Development+Testing, UAT).
- Billing: Monthly invoices based on timesheets showing hours per role, approved by Mr. Karthik before payment.

Aspect	Fixed Bid	Billing
Budget Control	Perfect Rs 2 Cr hard limit	Overrun risk during testing
Requirement Stability	Agri platform specs clear post-RA	Payment gateway uncertainty
Timeline Pressure	Revenue deadline critical	Scope creep delays launch
Client Involvement	Moderate (milestone approvals)	High (weekly sprint planning)
Vendor Incentive	Efficiency/profit within budget	More hours = more revenue

Q14. Preparer Timesheets of a BA in various stages of SDLC

- Design Timesheet of a BA
- Development Timesheet of a BA
- Testing Timesheet of a BA
- UAT Timesheet of a BA
- Deployment n Implementation Timesheet of a BA

DESIGN TIMESHEET

Design Timesheet					
S. No	Tasks	Actionable Items	Start Time	End Time	Durati on
1	Stakeholder review meeting - Design phase kickoff	Meeting with Peter, Kevin, Ben to review HLD approach	10:00 AM	11:00 AM	1 hour
2	Architecture walkthrough - Payment gateway integration	Technical discussion with Mr. Karthik on payment flow design	11:30 AM	1:00 PM	1.5 hours
3	Requirement traceability review	Mapping business requirements to design components	2:00 PM	3:30 PM	1.5 hours
4	Screen wireframe validation	Farmer UI/UX review for product browsing & ordering	3:45 PM	5:15 PM	1.5 hours
5	Design document finalization meeting	Team discussion on design approvals & sign-offs	5:30 PM	7:00 PM	1.5 hours
				Total	7.5 hours

DEVELOPMENT TIMESHEET

Development Timesheet					
S. No	Tasks	Actionable Items	Start Time	End Time	Durati on
1	Daily standup - Product catalog module	Developer clarifications on search/filter requirements	10:00 AM	10:30 AM	0.5 hours
2	Code walkthrough - Order processing logic	Review code implementation against functional specifications	10:45 AM	12:30 PM	1.75 hours
3	Requirement clarification - Payment integration	In-depth discussion with development team on edge cases	1:30 PM	3:00 PM	1.5 hours
4	RTM updates & traceability mapping	Update Requirement Traceability Matrix with dev completion status	3:15 PM	4:45 PM	1.5 hours
5	Feature handover documentation	Prepare handover notes for testing phase	5:00 PM	6:30 PM	1.5 hours

	Total	6.75 hours
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TESTING TIMESHEET

Testing Timesheet					
S. No	Tasks	Actionable Items	Start Time	End Time	Duration
1	Test case derivation meeting	Extract test cases from "Browse Products" & "Place Order" use cases	10:00 AM	11:30 AM	1.5 hours
2	SIT defect triage session	Review & prioritize defects with testing team & Mr. Dooku	11:45 AM	1:30 PM	1.75 hours
3	Requirement clarification for test scenarios	Discussion on edge cases & validation rules for payment	2:30 PM	4:00 PM	1.5 hours
4	Defect analysis & root cause discussion	Technical deep-dive on failed test cases with developers	4:15 PM	5:45 PM	1.5 hours
5	SIT completion & signoff documentation	Prepare SIT completion report & confirmation	6:00 PM	7:00 PM	1 hour
				Total	7.75 hours

UAT TIMESHEET

UAT Timesheet					
S. No	Tasks	Actionable Items	Start Time	End Time	Duration
1	UAT kickoff meeting with farmers	Meeting with Peter, Kevin, Ben to finalize UAT test scripts	10:00 AM	11:30 AM	1.5 hours
2	Business scenario walk-through	Explain real-world farmer workflows & expected system behavior	12:00 PM	1:30 PM	1.5 hours
3	UAT defect logging & prioritization	Document & triage 25+ UAT issues with severity levels	2:30 PM	4:30 PM	2 hours
4	Stakeholder communication - Defect updates	Zoom call to update farmers on defect resolution status	4:45 PM	5:45 PM	1 hour
5	Final UAT sign-off meeting	Completion walkthrough with Mr. Henry & stakeholders	6:00 PM	7:00 PM	1 hour
				Total	7.5 hours

DEPLOYMENT & IMPLEMENTATION TIMESHEET

Deployment and implementation Timesheet					
S. No	Tasks	Actionable Items	Start Time	End Time	Durati on
1	Production deployment planning meeting	Coordinate deployment checklist with DevOps & project team	10:00 AM	11:00 AM	1 hour
2	Farmer training material finalization	Review & approve training documentation for end-users	11:30 AM	1:00 PM	1.5 hours
3	Go-live support coordination	Monitor system performance & farmer feedback during launch	2:00 PM	5:00 PM	3 hours
4	Hypercare defect support	Triage & prioritize production issues with development team	5:15 PM	6:45 PM	1.5 hours
5	Project closure & lessons learned	RTM closure verification & final project documentation	7:00 PM	8:00 PM	1 hour
				Total	8 hours